

Mathematics for Social Scientists

Part 2: Introduction to Analysis Optimization

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Outline

Optimization

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Why optimization (for social scientists)?

- **Political science:** Parties choose platforms; governments allocate budgets; voters choose turnout to maximize utility.
- **Sociology:** Diffusion models choose parameters that best fit adoption curves.
- **Psychology:** Learning curves maximize performance for a given effort/time.

We use calculus to find *best* (max/min) choices of x by looking at slopes (derivatives) and curvature (second derivatives) — the same slope intuition you built in the derivatives unit.

Why optimization (in Statistics & ML)?

- **Ordinary Least Squares (OLS):** $\min_{\beta} \sum_{i=1}^n (y_i - x_i^{\top} \beta)^2$
- **Maximum Likelihood (MLE):** $\max_{\theta} \sum_{i=1}^n \log p(y_i \mid x_i; \theta)$
- **Regularization (bias-variance trade-off):**
Ridge: $\min_{\beta} \sum_i (y_i - x_i^{\top} \beta)^2 + \lambda \|\beta\|_2^2$ Lasso: $\min_{\beta} \sum_i (y_i - x_i^{\top} \beta)^2 + \lambda \|\beta\|_1$
- **Clustering:** k -means $\min_{\{\mathcal{C}_k\}} \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - \mu_k\|^2$

These show up across political science, sociology, and psychology when fitting models and predicting outcomes.

Maxima, Minima, Root Finding

Some Definitions

Definition 1 (Monotonically increasing).

A function $f(x)$ is called *monotonically increasing* (decreasing) if whenever $x \geq y$, we have $f(x) \geq f(y)$ ($f(x) \leq f(y)$).

Definition 2 (Convex/Concave Function).

Let f be a function that is twice differentiable (its second order derivative exists.) Then, it is called **convex** (concave) if it's second derivative is positive (negative).

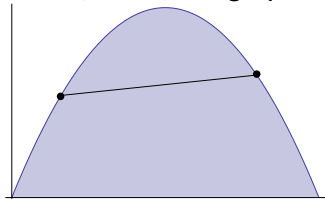
Remark: If a function is convex, then it means the line segment you get from connecting two points of the function line lies above the function. If it is concave, it lies below the function.

Ex: $f(x) = x^2$ is convex.

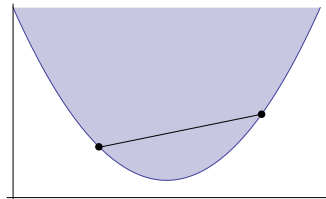
Ex: $f(x) = \ln(x)$ is concave.

More General Definition of Concavity/Convexity

A function is called concave (convex) if the line segment joining any two points on the graph is below (above) the graph, or on the graph.



concave



convex

We can derive the concavity/convexity of functions from the concept of convex sets.

Definition 3 (Critical Point).

The *critical points* of $f(x)$ are the points where the first derivative of the function is zero.

Definition 4 (Saddle point).

A **critical** point that is *not* a local maximum or minimum (i.e., not an extreme point).

Definition 5 (Inflection point).

A point where curvature changes (concave $\leftrightarrow$ convex); equivalently, f'' has opposite signs immediately to the left and right of the point. The slope there may be zero or nonzero.

Examples. $f(x) = x^3$: $x = 0$ is a *saddle* (critical, not an extreme; also an inflection).
 $f(x) = x^2 + 2x + 16$: $x = -1$ is a *critical* point (classify via f'').

Local vs. Global Extremes – intuition

Definition 6 (Local maximum / minimum).

A point a is a **local maximum** of f if, after you *zoom in* on a (look very close to a), the value $f(a)$ is higher than the values $f(x)$ at all nearby x .

A point a is a **local minimum** if, when you zoom in on a , $f(a)$ is lower than the values $f(x)$ at all nearby x .

Remark (Global version).

Global maximum/minimum means $f(a)$ is the highest/lowest value on the *entire* domain (not just nearby).

Example.

$f(x) = -(x - 1)^2$ has a local (and global) maximum at $x = 1$. $g(x) = x^3$ has no local max/min at $x = 0$ because values on the left and right keep getting higher/lower as you move away.

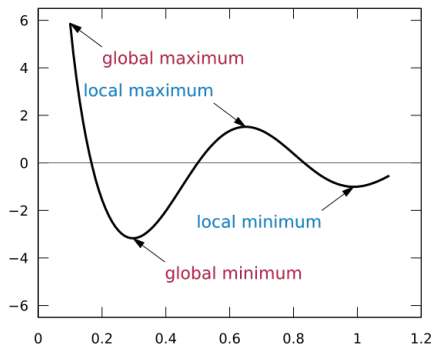
Finding extrema of a function

Definition 7.

Let $f : I \rightarrow \mathbb{R}$ be a function. f has a **local maximum** (**minimum**) at $a \in I$ if $\exists \varepsilon > 0$ such that

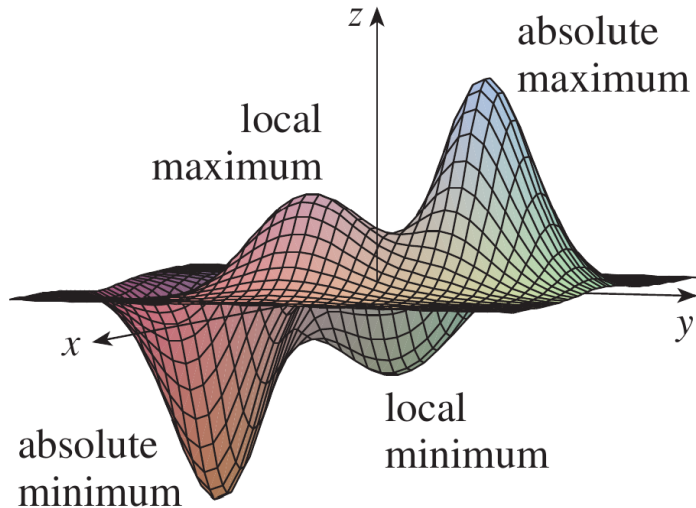
$$f(x) \leq f(a) \quad \forall x \in U \cap I. \quad (f(x) \geq f(a))$$

where $U = (a - \varepsilon, a + \varepsilon)$.



Finding extrema of a function

A function of two variables



Theorem 1.

Let $f : I \rightarrow \mathbb{R}$ be a function which is differentiable at $a \in I$. If a is a local maximum or minimum of f , then

$$f'(a) = 0.$$

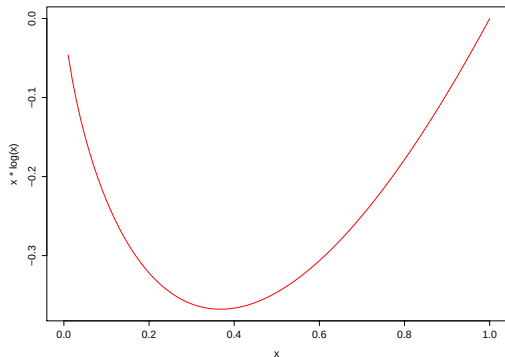


Figure: $f(x) = x \cdot \log(x)$

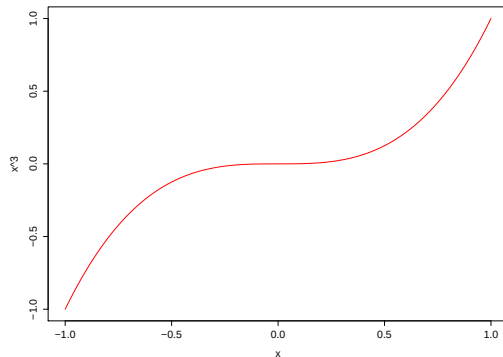


Figure: $f(x) = x^3$

Let $f : I \rightarrow \mathbb{R}$ be differentiable at $a \in I$ with $f'(a) = 0$. Then three cases are possible:

(i) $f''(a) > 0 \quad \Rightarrow f$ has a local minimum at a

(ii) $f''(a) < 0 \quad \Rightarrow f$ has a local maximum at a

(iii) $f''(a) = 0 \quad \Rightarrow f$ has a minimum or maximum or saddle point at a

Unconstrained Optimization

Unconstrained optimization: the basic recipe

- **Goal:** find extreme points (max/min) of f on its domain.
- **Step 1 (FOC) – candidates (critical points):** solve $f'(x) = 0$; also check where f' fails and any endpoints.
- **Step 2 (SOC) – classify (second derivative):** $f'' > 0$ min; $f'' < 0$ max; $f'' = 0$ inconclusive (could be a **saddle**).
- **Step 3 – global check:** compare f at all candidates and endpoints.

Motivation for Optimization

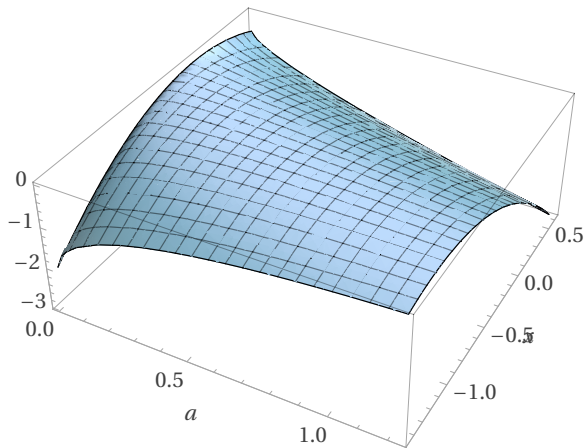
In decision theory, we are interested in the decision-making process of an individual.

Let us assume, we have a specified utility function of a person $u(x) = -(x + \sqrt{a})^2$.

We want to know the optimal choice the person can take. How do we do this?

Motivation for Optimization

$$u(x) = -(x + \sqrt{a})^2.$$



Computed by Wolfram|Alpha

Single Variable Optimization – FOC

The first step is the **first-order condition (FOC)**:

$$\frac{df}{dx} \equiv 0$$

- At an interior extreme point, the slope (first derivative) is zero – this gives **critical points** (candidates).
- In our example $u(x) = -(x + \sqrt{a})^2$: $\frac{du}{dx} = -2x - 2\sqrt{a}$.
- Solving $\frac{du}{dx} = 0$ gives $x^* = -\sqrt{a}$.
- This is a candidate; it could be a local maximum, a local minimum, or a saddle (critical but not extreme).

Single Variable Optimization – SOC

Now we decide which case applies by checking the **second-order condition**.

- Local maximum if $\frac{d^2f}{dx^2}(x^*) < 0$ (concave at x^*).
- Local minimum if $\frac{d^2f}{dx^2}(x^*) > 0$ (convex at x^*).
- If $\frac{d^2f}{dx^2}(x^*) = 0$, the test does not decide. In our convention: if the first nonzero higher derivative is of *odd* order (e.g. $f^{(3)}(x^*) \neq 0$), then x^* is a **saddle** (critical, not an extreme).

For $u(x) = -(x + \sqrt{a})^2$, we have $u''(x) = -2$, so at $x^* = -\sqrt{a}$ there is a **local maximum** (indeed global here).

Bivariate Optimization I

Consider a C^2 function (i.e. continuous with continuous first/second partials) $f(x, y)$ on a convex domain S (the domain; i.e. the line segment between any two points of S lies in S).

First-order condition

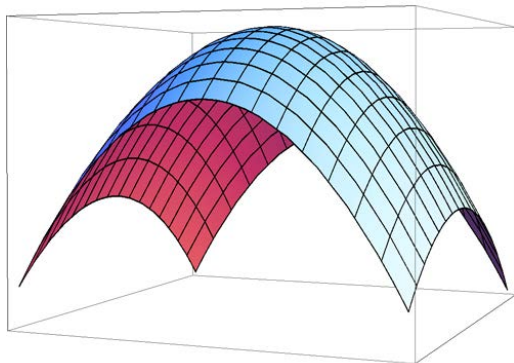
- Find the first-order partial derivatives and set them to zero: $f_x(x, y) = 0$, $f_y(x, y) = 0$.
- Solve for (x, y) ; any solution (x^*, y^*) is a **critical point**.

Second-order condition

- *Local maximum near (x^*, y^*) :* if $f_{xx} \leq 0$, $f_{yy} \leq 0$, $f_{xx}f_{yy} - (f_{xy})^2 \geq 0$.
- *Local minimum near (x^*, y^*) :* if $f_{xx} \geq 0$, $f_{yy} \geq 0$, $f_{xx}f_{yy} - (f_{xy})^2 \geq 0$.
- If the last expression is < 0 , we have a **saddle** (not a max/min); if $= 0$, the test is inconclusive.

Bivariate Optimization II

Consider the function $f(x, y) = -\frac{1}{2}(x - 1)^2 - y^2$.



Bivariate Optimization III

Function $f(x, y) = -0.5(x - 1)^2 - y^2$.

The first order condition

$$\frac{\partial f}{\partial x} = -x + 1 \equiv 0$$

$$\frac{\partial f}{\partial y} = -2y \equiv 0$$

gives us a **critical point** at $x = 1, y = 0$.

Bivariate Optimization IV

The second order condition

$$\frac{\partial^2 f}{\partial x^2} = -1 < 0$$

$$\frac{\partial^2 f}{\partial y^2} = -2 < 0$$

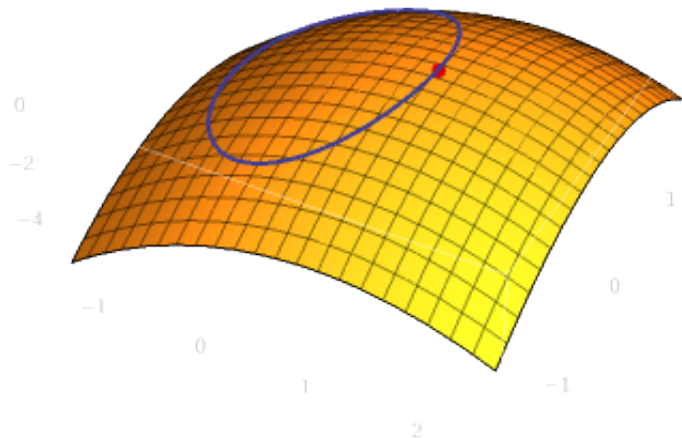
$$\frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (-1) \cdot (-2) - 0 \geq 0$$

tells us that we have a **local maximum** at $x = 1, y = 0$.

Constrained Optimization

Constrained Optimization - Graphically

Consider the function $f(x, y) = -\frac{1}{2}(x - 1)^2 - y^2$, subject to $x^2 + y^2 = 1$



Optimization Under Constraints - Problem

So far we have considered decision problems in general. But what about situations in which an agent has to make her decision under given constraints?

Let us consider the following example:

We as a city can decide to allocate our budget between cultural (c) and social (s) affairs.

The overall utility function of our city is given by $f(c, s) = \frac{1}{2}s^2 + \left(c - \frac{1}{3}\right)^2$. Our budget is constrained as $c + s = 2$.

A method to solve such problems is the so-called **Lagrangian multiplier method**.

Lagrangian Multiplier Method I

In order to solve the maximization problem $\max f(x, y)$ subject to $g(x, y) = c$ we proceed the following way.

1. Write down the Lagrangian $\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c)$, where λ is a constant.
2. Differentiate $\mathcal{L}$ with respect to x and y , and equate the partial derivatives to 0.
3. Solve the system of equations that the two partials form together with the constraint.

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x} &= \frac{\partial f}{\partial x} - \lambda \frac{\partial g}{\partial x} \equiv 0 \\ \frac{\partial \mathcal{L}}{\partial y} &= \frac{\partial f}{\partial y} - \lambda \frac{\partial g}{\partial y} \equiv 0 \\ g(x, y) &= c\end{aligned}$$

Application to our problem

The Lagrangian

$$\mathcal{L}(c, s, \lambda) = \frac{1}{2}s^2 + \left(c - \frac{1}{3}\right)^2 - \lambda(s + c - 2).$$

The system of equations

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial s} &= s - \lambda \equiv 0 \\ \frac{\partial \mathcal{L}}{\partial c} &= 2c - \frac{2}{3} - \lambda \equiv 0 \\ s + c &= 2\end{aligned}$$

If we solve the system of equations, we get $c = \frac{8}{9}$ and $s = \frac{10}{9}$ (and $\lambda = \frac{10}{9}$).

Lagrangian Multiplier Method II

If we compare the Lagrangian method for constrained optimization to the unconstrained optimization, is still something missing?

Yes, theoretically we have to check for the second-order condition.

You find the formulation in Sysdsæter/Hammond (2008) on pp. 506–507.

Advanced Constrained Optimization

There is much more to constrained optimization!

- Multivariable optimization (we need matrix algebra for that!).
- Lagrangian for more than two variables.
- Lagrangian for more than one condition.
- Optimization for inequalities
 $\max f(x, y)$ subject to $g(x, y) \leq c$
“nonlinear programming” or “Kuhn–Tucker”

See Sysdsæter/Hammond (2008), Chapter 14.

Appendix

Gradient Vector (many inputs)

Definition 8 (Gradient).

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the *gradient* at x is the column vector of first partials

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \frac{\partial f}{\partial x_2}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}.$$

Critical points solve $\nabla f(x^*) = \mathbf{0}$. (Same “FOC” idea as the single-input case.)

This is the multi-input version of “set the derivative to zero.”

Hessian Matrix (curvature in many inputs)

Definition 9 (Hessian).

The *Hessian* of f at x collects all second partial derivatives into a $n \times n$ matrix

$$H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.$$

- At a critical point x^* , $H_f(x^*)$ classifies the point: positive definite $\Rightarrow$ local min; negative definite $\Rightarrow$ local max; indefinite $\Rightarrow$ saddle; semidefinite $\Rightarrow$ inconclusive without more info.
- You can also view $H_f(x^*)$ as the curvature term in the second-order Taylor approximation around x^* .

Quick 2-input classification (at a critical point)

Let $f_x = f_y = 0$ at (x^*, y^*) . Define

$$D = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - \left(f_{xy}(x^*, y^*) \right)^2.$$

Then

$$\begin{cases} D > 0, f_{xx} > 0 \Rightarrow \text{local minimum,} \\ D > 0, f_{xx} < 0 \Rightarrow \text{local maximum,} \\ D < 0 \Rightarrow \text{saddle (not a max/min),} \\ D = 0 \Rightarrow \text{test inconclusive.} \end{cases}$$

(This is the familiar 2×2 Hessian test written in second-partial form.)

Extreme Value (Weierstrass) Theorem

Theorem 2 (Weierstrass/Extreme Value Theorem).

If f is continuous on a nonempty, closed and bounded (i.e. compact) set $S \subset \mathbb{R}^n$, then there exist points $c, d \in S$ with

$$f(d) \leq f(x) \leq f(c) \quad \text{for all } x \in S.$$

Intuition: on a compact domain you *must* hit a global min and max—useful when searching for optima on boxes/closed intervals.

Comparative statics (why parameters matter)

Question: how does the optimizer $x^*(\theta)$ move when a parameter θ changes?

- In simple problems, implicitly differentiate the FOC(s) to get $\frac{dx^*}{d\theta}$.
- Example you already use: for $u(x) = -(x + \sqrt{a})^2$, the solution is $x^*(a) = -\sqrt{a}$ and $\frac{dx^*}{da} = -\frac{1}{2\sqrt{a}}$ (sign tells you direction of change).

Small parameter changes $\Rightarrow$ predictable shifts in the optimal choice—this is the backbone of many testable predictions.

Sets & Topology – primer

Definition 10 (Interior / Boundary).

Interior point: some ε -ball around $\mathbf{x}$ lies entirely in S . Boundary point: every ε -ball around $\mathbf{x}$ meets S and its outside. *Example:* for $x^2 + y^2 \leq 4$, interior is $x^2 + y^2 < 4$, boundary is $x^2 + y^2 = 4$.

Definition 11 (Open / Closed).

Open: all points are interior (edge not included). Closed: contains all boundary points (edge included).
Examples: $x^2 + y^2 < 4$ open; $x^2 + y^2 \leq 4$ closed; $2 < x^2 + y^2 \leq 4$ neither.

Definition 12 (Closure).

The smallest closed set containing S (i.e., S plus any missing boundary). *Example:*
 $\overline{\{x^2 + y^2 < 4\}} = \{x^2 + y^2 \leq 4\}.$

Sets & Topology – primer

Definition 13 (Bounded).

S fits inside some ball of radius R . *Examples:* $[a, b]$, a disk of radius 2. *Counterexample:* $(-\infty, 0] \cup [1, \infty)$ (closed but unbounded).

Definition 14 (Compact (in $\mathbb{R}^n$)).

Closed *and* bounded. *Why care:* continuous f on a nonempty compact set attains a global max and min (Weierstrass).

Definition 15 (Convex).

For any $x, y \in S$ and $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda)y$ is in S (the straight line stays inside). *Examples:* intervals, rectangles/boxes, disks; nonconvex: a ring/donut.